

SUN Chen

Research Fellow

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RESEARCH PROFILE

My research focuses on back-end-of-line (BEOL)-compatible oxide-semiconductor (OS) devices for monolithic 3D (M3D) integration, with emphasis on high-density non-volatile memory (NVM) and compute-in-memory (CiM) applications. Current work includes OS ferroelectric field-effect transistors (FeFETs), 3D ferroelectric vertical NAND, 2T0C DRAM, and physics-based reliability and variability studies using advanced electrical characterization, TCAD, and compact modeling.

RESEARCH HIGHLIGHTS

- **First-author/co-first-author and corresponding/co-corresponding-author publications**
 - 10 journal papers in IEEE Electron Device Letters (EDL), IEEE Transactions on Electron Devices (TED), and Advanced Electronic Materials
 - 10 conference papers, **including 5 at International Electron Devices Meeting (IEDM) and 3 at VLSI Symposium (two flagship conferences for semiconductor devices and microelectronics)**
- **Overall publications**
 - 43 journal publications, mainly in IEEE EDL and IEEE TED
 - 52 conference publications, including 19 papers at IEDM and 24 papers at the VLSI Symposium
- **Award-Winning Papers:** 2025 Symposium on VLSI Technology Best Demo Paper, 2024 and 2021 Symposium on VLSI Technology Best Student Paper, and 2019 ICICDT Best Paper
- Co-inventor of one granted U.S. patent
- h-index: 22; citations: 1500 ([Google Scholar](#), as of Aug. 2026)

RESEARCH INTERESTS

- **Current Research: Emerging Semiconductor Devices, Modeling, and CiM Applications**
 - Developing high-performance oxide-semiconductor (OS, e.g., IGZO, ZnO, In₂O₃) FeFETs and high-density 3D ferroelectric vertical NAND for monolithic 3D integration
 - Establishing physics-based and TCAD-calibrated models to understand operation mechanisms, reliability, and variability in emerging OS ferroelectric devices
 - Exploring energy-efficient and high-density compute-in-memory applications
- **Future Research Vision: Cross-Layer Co-Design (Device-Array-System)**
 - Device-to-Array Synergy: Architecting array-level scaling of BEOL-compatible non-volatile memories leveraging OS FeFETs and 3D ferroelectric vertical NAND
 - High-Bandwidth Flash (HBF): Pioneering HBF utilizing M3D-stacked OS ferroelectric vertical NAND arrays to bridge the memory-bandwidth gap
 - System Reliability: Establishing physics-based modeling frameworks to mitigate device degradation and thermal coupling in ultra-high-density M3D systems

APPOINTMENTS

- **Research Fellow** Feb. 2023 – Present
 - National University of Singapore, Electrical and Computer Engineering
 - Lead device fabrication, electrical characterization, and reliability studies, supported by TCAD and compact-model analysis
 - Contribute to proposal development and technical planning for funded projects
 - Mentor Ph.D. and master's students
- **Software Engineer** Jul. 2018 – Dec. 2018
 - Chengdu Institute, Huawei Technologies Co., Ltd.

EDUCATION

- **Ph.D.** Jan. 2019 – Jan. 2023
 - Supervisor: Gong Xiao, National University of Singapore, Electrical and Computer Engineering
 - Dissertation: “Amorphous Indium-Gallium-Zinc-Oxide-Based Transistors for Emerging Non-Volatile Memory and Neuromorphic Computing Applications”
- **B. Eng.** Sep. 2014 – Jun. 2018
 - University of Electronic Science and Technology of China (UESTC)
 - Electronic Information Engineering (Ranking: Top 1%; GPA: 3.91/4.00, 90.06/100)

INTERNATIONAL COLLABORATIONS

- **Academic Collaborations**
 - University of Notre Dame | 2021 – Present: Oxide-semiconductor FeFETs for high-density 3D ferroelectric vertical NAND.
 - Zhejiang University and Rochester Institute of Technology (RIT) | 2020 – Present: BEOL-compatible ternary content-addressable memory using IGZO-based FeFETs and floating-gate transistors.
 - Tsinghua University | 2021 – 2024: Oxide-semiconductor TFT-based DRAM for low-power and high-retention charge-domain computing.
- **Industry Collaborations**
 - Soitec, STMicroelectronics, CEA-LETI, and IMEC | 2019 – 2022: SSOI FinFETs for high-performance logic and RF applications.

CONTRIBUTIONS TO FUNDED RESEARCH PROJECTS

- **Competitive Research Grants (Singapore MOE)**
 - Enabling Extremely-Low Power AI Hardware Using Novel Ferroelectric Capacitive Synapse and Arrays | Aug. 2025 – Present
 - Key contributions: supporting project execution and experimental coordination; mentoring two Ph.D. students; leading device design, fabrication, and reliability analysis
 - Enabling Photonic Reservoir Computing Using Monolithic Opto-Electronic Integrated Circuits | Mar. 2023 – Mar. 2026
 - Key contributions: mentored one Ph.D. student; led device layout design, process integration, and fabrication
 - Advanced Steep Slope Nano-electronic Devices Towards Sub-0.3 V Supply Voltage | Jul. 2019 – Jul. 2022
 - Key contributions: physics-based modeling, data interpretation, and technical reporting
- **Industry-Sponsored Research**
 - SMART-Cut Technology for Advanced System-on-Chip (Soitec) | May 2022 – Present
 - Key contributions: conducting electrical characterization and TCAD-assisted simulation; coordinating technical discussions and reporting with industry partners
- **Internal Research Projects (NUS)**
 - Amorphous-Oxide-Semiconductor Thin Film Transistors and DRAM to Enable 3D Monolithically Integrated Architecture for Near/In-memory Computing | Oct. 2020 – Oct. 2023
 - Key contributions: supported proposal writing and coordinated project experiments; mentored two Ph.D. students; led device fabrication, electrical characterization, and data analysis

MENTORING & TEACHING EXPERIENCE

- Mentoring Ph.D. and master's students in research projects on oxide-semiconductor devices, ferroelectric memories, associative computing, and opto-electronic devices
 - Guided students on device physics, performance, and reliability mechanisms (carrier mobility, memory window, endurance, retention, bias-stress instability, variability)

- Co-developed material stacks, device concepts, and experimental plans, including structure design and process-window optimization under thermal and integration constraints
- Supervised fabrication flows (ALD thin films, lithography, etch/lift-off, anneal schemes) and coordinated cross-tool runs to ensure process repeatability
- Led electrical characterization and data analysis (I_D – V_G / I_D – V_D , pulsed/AC tests, endurance/retention, noise) and physics-based interpretation
- Directed device modeling and simulation using TCAD and compact models to correlate materials, process, and structural variations with electrical behavior and guide design iterations
- **Mentored students in manuscript preparation and conference submissions, contributing to multiple IEEE EDL, IEDM, VLSI Symposium, and ICCAD papers**
- Undergraduate Teaching — Teaching Assistant for the NUS Bachelor of Technology (BTech) programme; topic: ion implantation simulation using the Stopping and Range of Ions in Matter (SRIM) software; class size: 50 students
 - Delivered lectures to the class
 - Developed and revised lab assignments
 - Conducted tutorial and Q&A sessions

TEACHING INTERESTS

- Semiconductor fundamentals and device physics
- Semiconductor devices, processing, and fabrication
- Advanced semiconductor devices and emerging memory technologies
- Device reliability, modeling, and compute-in-memory technologies

PROFESSIONAL SERVICE

- **Peer Reviewer**
 - IEEE Electron Device Letters, IEEE Transactions on Electron Devices, IEEE Journal of the Electron Devices Society, Advanced Functional Materials, Advanced Science, Advanced Intelligent Systems, and others.

ACADEMIC AWARDS

- 2025 IEEE Symposium on VLSI Technology, Best Demo Paper Award (co-author)
- 2024 IEEE Symposium on VLSI Technology, Best Student Paper Award (co-author)
- 2021 IEEE Symposium on VLSI Technology, Best Student Paper Award (co-author)
- 2019 IEEE International Conference on IC Design and Technology, Best Paper Award (first author)

SELECTED PUBLICATIONS

- **High-Performance Oxide-Semiconductor Devices and 3D Ferroelectric Memory**
- 1. H. Liu, Y. Le, W. Shi, X. Li, G. Liu, D. Zhang, R. Shao, **C. Sun[#]**, and X. Gong[#], “Strain-induced mobility enhancement in crystalline In_2O_3 FETs,” *IEEE Electron Device Letters*, vol. 47, no. 8, pp. 1603-1606, Aug. 2026. **(co-corresponding author)**
- 2. X. Li, N. Ji, L. Jiao, Y. Tong, G. Liu, C. Wu, T. D. Hieu, Z. Zhang, J. Sun, K. Ni, G. Liang, **C. Sun[#]**, D. Zhang[#], and X. Gong[#], “BEOL-compatible oxide semiconductor twin-gate FeFET achieving a record-high memory window of 25.4 V and a near-ideal memory window-to-operation voltage ratio of 1.88,” in 2026 IEEE/JSAP Symposium on **VLSI Technology and Circuits**, June 2026, pp. 1–3. **(co-corresponding author)**
- 3. **C. Sun**, X. Li, D. Zhang, W. Shi, Y. Chen, G. Liu, R. Shao, Q. Kong, Y. Wang, L. Jiao, Z. Zhou, Y. Feng, X. Wang, K. Ni, and X. Gong, “First demonstration of 3D Fe-VNAND with function-separated dual-layer oxide semiconductor channel and laminated ferroelectric stack,” in *2025 International Electron Devices Meeting (IEDM)*, Dec. 2025.

4. **C. Sun**, K. Han, S. Samanta, Q. Kong, J. Zhang, H. Xu, X. Wang, A. Kumar, C. Wang, Z. Zheng, X. Yin, K. Ni, and X. Gong, “Highly scaled InGaZnO ferroelectric field-effect transistors and ternary content-addressable memory,” *IEEE Transactions on Electron Devices*, vol. 69, no. 9, pp. 5262–5269, Sep. 2022.
5. **C. Sun**, Z. Zheng, K. Han, S. Samanta, J. Zhou, Q. Kong, J. Zhang, H. Xu, A. Kumar, C. Wang, and X. Gong, “Temperature-dependent operation of InGaZnO ferroelectric thin-film transistors with a metal-ferroelectric-metal-insulator-semiconductor structure,” *IEEE Electron Device Letters*, vol. 42, no. 12, pp. 1786–1789, Dec. 2021.
6. **C. Sun**, K. Han, S. Samanta, Q. Kong, J. Zhang, H. Xu, X. Wang, A. Kumar, C. Wang, Z. Zheng, X. Yin, K. Ni, and X. Gong, “First demonstration of BEOL-compatible ferroelectric TCAM featuring a-IGZO Fe-TFTs with large memory window of 2.9 V, scaled channel length of 40 nm, and high endurance of 10^8 cycles,” in *2021 Symposium on VLSI Technology*, Jun. 2021, pp. 1–2.

• **Physics-Based Comprehensive Modeling: Mechanisms, Reliability, and Variability**

7. W. Zhang*, **C. Sun***, X. Li, Z. Wang, J. Wang, Y. Cen, D. Zhang, T. Du, Y. Lu, X. Gong#, and X. Fong#, “Multi-grain ferroelectric induced drain current variability in BEOL-compatible OS FeFETs,” *IEEE Electron Device Letters*, vol. 47, no. 3, pp. 506–509, Mar. 2026. **(co-first/corresponding author)**
8. **C. Sun**, Q. Kong, G. Liu, D. Zhang, L. Jiao, X. Wang, J. Zhang, H. Xu, Y. Feng, R. Shao, Y. Chen, and X. Gong, “Comprehensive investigation of bias stress-induced instabilities in highly scaled ZnO FeFETs: Impact of channel thickness, channel length, and switching cycles,” *IEEE Transactions on Electron Devices*, vol. 72, no. 5, pp. 2334–2340, May 2025.
9. **C. Sun**, Q. Kong, G. Liu, D. Zhang, L. Jiao, X. Wang, J. Zhang, H. Xu, Y. Feng, R. Shao, Y. Chen, and X. Gong, “Understanding bias stress-induced instabilities in ALD-deposited ZnO FeFETs featuring HZO-Al₂O₃-HZO ferroelectric stack,” *IEEE Electron Device Letters*, vol. 45, no. 11, pp. 2122–2125, Nov. 2024.
10. X. Wang*, **C. Sun***, Z. Zheng, L. Jiao, Z. Zhou, D. Zhang, G. Liu, and X. Gong, “Comprehensive experiments and modeling applicable for ferroelectric transistors with an MFMIS structure and a wide range of area ratios: Unveiling the operation mechanisms,” *IEEE Transactions on Electron Devices*, vol. 71, no. 9, pp. 5332–5338, Sep. 2024. **(co-first author)**
11. X. Wang*, **C. Sun***, Z. Zheng, L. Jiao, Z. Zhou, D. Zhang, G. Liu, Q. Kong, J. Zhang, H. Xu, K. Han, Y. Kang, L. Liu, and X. Gong, “Deep insights into the interplay of polarization switching, charge trapping, and soft breakdown in metal-ferroelectric-metal-insulator-semiconductor structure: Experiment and modeling,” in *2022 IEEE International Electron Devices Meeting (IEDM)*, Dec. 2022, p. 13.3.1-13.3.4. **(co-first author)**
12. **C. Sun**, K. Han, and X. Gong, “Performance evaluation of static random access memory (SRAM) based on negative capacitance FinFET,” in *2019 International Conference on IC Design and Technology (ICICDT)*, Jun. 2019, pp. 1–4.

• **Compute-in-Memory Applications**

13. Y. Tong, J. Lu, Y. Le, K. Ni, **C. Sun**#, and X. Gong#, “TFCM-CAM: A charge-domain CAM based on highly scaled trench ferroelectric capacitive memory for dense and energy-efficient associative computing,” to appear in *2026 IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, Nov. 2026. **(co-corresponding author)**
14. R. Shao, Y. Chen, Y. Wang, X. Chen, X. Jia, J. Sun, W. Wang, **C. Sun**#, and X. Gong#, “First demonstration of monolithic 3D integrated Si PICs and BEOL oxide semiconductor EICs for electronic-photonic compute-in-memory (CiM),” in *2025 International Electron Devices Meeting (IEDM)*, Dec. 2025. **(co-corresponding author)**
15. **C. Sun**, X. Wang, H. Xu, J. Zhang, Z. Zheng, Q. Kong, Y. Kang, K. Han, L. Jiao, Z. Zhou, Y. Chen, D. Zhang, G. Liu, L. Liu, and X. Gong, “Novel a-IGZO anti-ferroelectric FET LIF neuron with co-

integrated ferroelectric FET synapse for spiking neural networks,” in *2022 IEEE International Electron Devices Meeting (IEDM)*, Dec. 2022, p. 2.1.1-2.1.4.

16. **C. Sun**, C. Li, S. Samanta, K. Han, Z. Zheng, J. Zhang, Q. Kong, H. Xu, Z. Zhou, Y. Chen, C. Zhuo, K. Ni, X. Yin, and X. Gong, “Computational associative memory with amorphous metal-oxide channel 3D NAND-compatible floating-gate transistors,” *Advanced Electronic Materials*, vol. 8, no. 12, p. 2200643, Sep. 2022.
17. J. Liu*, **C. Sun***, W. Tang, Z. Zheng, Y. Liu, H. Yang, C. Jiang, K. Ni, X. Gong, and X. Li, “Low-power and scalable retention-enhanced IGZO TFT eDRAM-based charge-domain computing,” in *2021 IEEE International Electron Devices Meeting (IEDM)*, Dec. 2021, p. 21.1.1-21.1.4. (**co-first author**)

- **Prior Research Experience: Strained Silicon-on-Insulator CMOS Devices**

18. J. Liang*, **C. Sun***, H. Xu, E. Y.-J. Kong, B.-Y. Nguyen, W. Schwarzenbach, C. Maleville, R. Berthelon, O. Weber, F. Arnaud, A. V.-Y. Thean, and X. Gong, “Strained silicon-on-insulator platform for co-integration of logic and RF—Part II: Comb-like device architecture,” *IEEE Transactions on Electron Devices*, vol. 69, no. 4, pp. 1769–1775, Apr. 2022. (**co-first author**)
19. **C. Sun**, H. Xu, J. Liang, E. Y.-J. Kong, B.-Y. Nguyen, W. Schwarzenbach, C. Maleville, R. Berthelon, O. Weber, F. Arnaud, X. Wang, A. V.-Y. Thean, and X. Gong, “Strained silicon-on-insulator platform for co-integration of logic and RF—Part I: Implant-induced strain relaxation,” *IEEE Transactions on Electron Devices*, vol. 68, no. 4, pp. 1425–1431, Apr. 2021.
20. **C. Sun**, J. Liang, H. Xu, E. Kong, B.-Y. Nguyen, A. Vandooren, W. Schwarzenbach, C. Maleville, V. Barral, R. Berthelon, O. Weber, F. Arnaud, A. V. Y. Thean, and X. Gong, “Enabling UTBB strained SOI platform for co-integration of logic and RF: Implant-induced strain relaxation and comb-like device architecture,” in *2020 IEEE Symposium on VLSI Technology*, Jun. 2020, pp. 1–2.

PATENTS

- (**Granted US Patent**) B. Nguyen, C. Maleville, W. Schwarzenbach, G. Xiao, A. Thean, **C. Sun**, and H. Xu, “Device architectures with tensile and compressive strained substrates,” US12218201B2.